



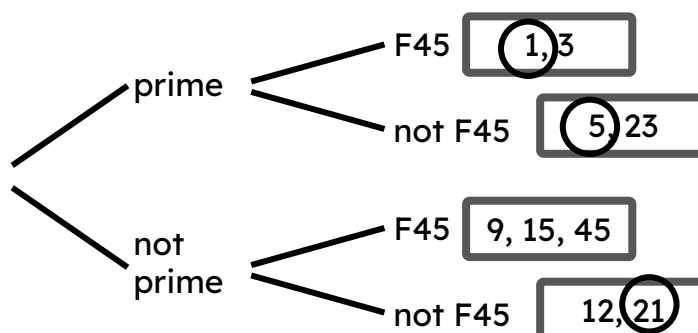
Comparing representations of outcomes for two events

Task A: Transferring information between diagrams

1) Jacob and Laura submit homework for showing the outcomes of two events from a trial using different representations.

Use Jacob's correctly written outcome table to circle any errors in the outcome tree and write down what is wrong with the outcome tree.

	factor of 45 (F45)	not factor of 45
prime	3, 5	7, 23
not prime	1, 9, 15, 45	12



1 is not a prime number.

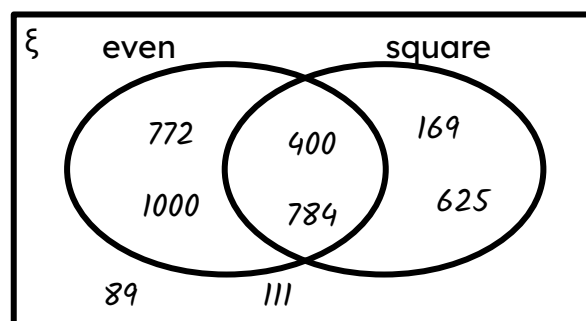
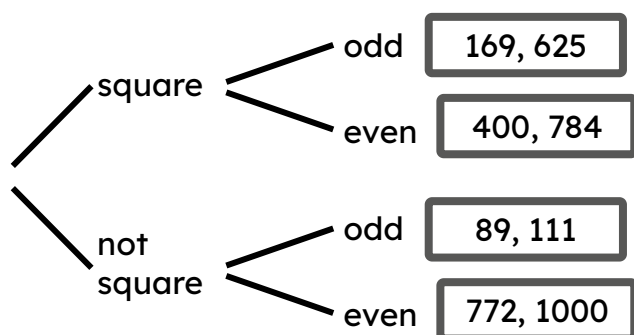
5 is also a factor of 45.

7 is missing - it should be a prime number that isn't a factor of 45.

21 should be removed, as it isn't an outcome of the trial.

2) This outcome tree correctly shows two events from the trial of spinning a spinner with 8 outcomes.

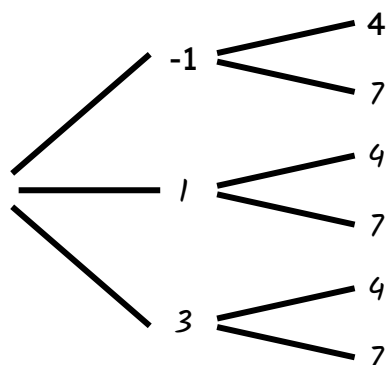
Use this outcome tree to complete the Venn diagram that shows the same trial.



3) The outcome tree and outcome table show the same two-stage trial: two spinners that will be spun (once each) with their outcomes added together.

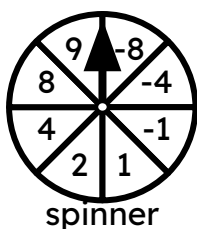
Both diagrams are accurate, but have some information missing.

The outcome table shows the sample space of outcomes after addition. Complete both diagrams.



+	-1	1	3
4	3	5	7
7	6	8	10

Task B: Using multiple diagrams



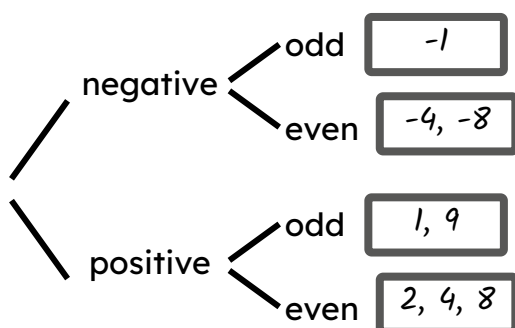
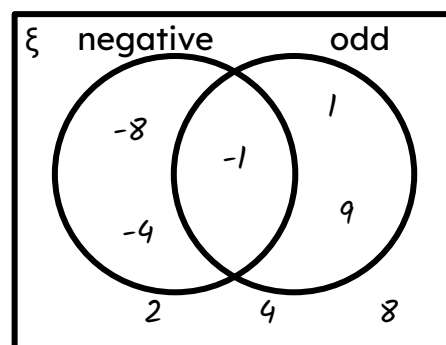
spinner

1) Jacob will spin this spinner and consider the events that the spinner will land on a negative number and an odd number.

a) Complete each diagram for these two events.

b) Write a sentence explaining which diagram you prefer.

	negative	positive
odd	-1	1, 9
even	-4, -8	2, 4, 8



2) Sofia will choose a starter and a main course.

a) By choosing two different representations, write down the possible outcomes of Sofia's meal.

*These representations could have been chosen:
listing, outcome table, and
outcome tree*

Here's an example of an outcome table that represents this two-stage trial.

	<i>B</i>	<i>S</i>	<i>C</i>	<i>T</i>
<i>L</i>	<i>LB</i>	<i>LS</i>	<i>LC</i>	<i>LT</i>
<i>P</i>	<i>PB</i>	<i>PS</i>	<i>PC</i>	<i>PT</i>

Starters

Leek Soup
Prawn Toast

Mains

Beef Lasagne
Shepherd's Pie
Chicken Burger
Tomato Curry

b) Write a sentence explaining why you chose those two representations.

List - the number of possible combinations is small and lists are easy to write.

Outcome table/tree - makes it easy to see I have found all possible combinations.

